# When Does Genomics Add Predictive Value Beyond Longitudinal Clinical History?

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***Abstract*—Genomic risk and longitudinal clinical history are distinct information channels for forecasting future disease. We ask when a static germline genomic channel can add predictive value once a longitudinal clinical-history score is already informative. The central result is finite-stage boundary localization: at a fixed clinical stage, if $b_{\delta,n}$ is the $(1-\delta)$ control quantile of the residual genomic log-likelihood-ratio magnitude, then genomics can reverse a clinical likelihood-ratio decision outside a log-LR neighborhood of half-width $b_{\delta,n}$ around the decision boundary only on a control-side event of probability at most $\delta$. The result is finite-stage and requires only that the relevant residual-genomic quantile be finite at the clinical stage analyzed. We test the theory in a Colorado fibrotic interstitial lung disease analysis using ZeBRA and an array-derived genomic feature panel, while treating *MUC5B* rs35705950 separately as the focal mechanistic genomic channel. Broad genomic augmentation provides essentially no paired incremental FILD/FILA AUC after conditioning on ZeBRA, yet *MUC5B* remains locally informative in restricted ZeBRA regimes. A direct cross-fitted score-level likelihood-ratio analysis estimates stable 95% and 99% residual-genomic scales of approximately 0.87 and 0.91 log-LR units, respectively, and shows that genomic-induced decision flips remain concentrated near the clinical boundary. The resulting pattern is regime-dependent complementarity rather than global genomic redundancy.**

***Index Terms*—ZeBRA, fibrotic interstitial lung disease, idiopathic pulmonary fibrosis, MUC5B, genomics, likelihood ratio, Neyman–Pearson lemma, Bayesian inference, decision boundary.**

## I. Motivating problem: ZeBRA and genomic risk in fibrotic ILD

Fibrotic ILDs, including idiopathic pulmonary fibrosis (IPF), have a strong but non-Mendelian genetic component. The *MUC5B* promoter variant rs35705950 is among the strongest common genetic risk factors for pulmonary fibrosis[1–3]. Its relevance extends beyond classic IPF: rs35705950 has been associated with interstitial lung abnormalities and definite CT evidence of pulmonary fibrosis in population cohorts, with progression of interstitial lung abnormalities, and with rheumatoid-arthritis-associated ILD[4–6]. At the same time, disease emergence is accompanied by a growing sequence of clinically observable diagnoses, prescriptions, procedures, and comorbidities. Longitudinal EHR histories have repeatedly been shown to support prediction of future diagnoses and clinical outcomes[7,8], and ZeBRA is designed to summarize such routinely observed histories into a future-risk score without requiring laboratory tests, imaging, or questionnaires[9].

Genetic information can provide substantial incremental prediction when the available clinical comparator is relatively weak. For IPF and interstitial lung abnormalities, Moll *et al.* reported an AUC of approximately 0.61 for a clinical model and 0.81 after combining that model with rs35705950 and an IPF polygenic risk score[10]. The question addressed here is therefore not whether genomics can ever improve clinical prediction, but how its incremental value changes once a longitudinal clinical-history channel is already strongly informative.

Our recent fibrotic ILD analyses ask whether genomic information improves prediction once ZeBRA is known. The most direct global test is a paired regularized comparison in which ZeBRA-only and ZeBRA-plus-genomics models are evaluated on the same held-out subjects. For FILD/FILA, the mean paired AUC increment from the broader array-derived genomic panel is only $+0.00043$, the median increment is 0, and the split-wise 2.5th–97.5th percentile range spans approximately $-0.0051$ to $+0.0041$; average precision decreases and both Brier score and log loss worsen after broad genomic augmentation. A separate nonlinear combined learner also fails to improve on ZeBRA, but that model-specific comparison is not itself a theoretical requirement. The broader panel construction and its distinction from the focal rs35705950 T-carrier channel are specified in Supplementary Table S1. This distinction between association and incremental discrimination is important: a marker can have a strong disease association without being a strong classifier, and changes in AUC can be small when the baseline model already discriminates well[11–13].

The absence of reproducible global gain is not explained by ZeBRA reconstructing the major genomic signal. Among subjects with called rs35705950 genotype, ZeBRA predicts *MUC5B* carrier status at chance level (AUC approximately 0.509). A modest pooled enrichment of carriers in the extreme ZeBRA tail disappears after stratification by FILD/FILA status, showing that the pooled association is consistent with both channels tracking disease rather than with clinical history encoding genotype. Yet residual *MUC5B* disease information is strongly heterogeneous across ZeBRA score space. This motivates the central question of the paper: how can a genomic channel remain useful for selected decisions while adding essentially nothing to global discrimination?

The apparent paradox is therefore:

> A strong genomic risk factor can be biologically important, and can improve decisions for selected individuals, while contributing almost nothing to global discrimination once a strong clinical-history predictor is available.

We address this question first at the level of likelihood ratios, because that formulation applies directly to both the theory and the empirical score-level analysis.

## II. Likelihood-ratio representation

For a fixed prediction horizon $h > 0$, let

$$Y = \mathbf{1}\{T \le t + h\} \in \{0, 1\}, \tag{1}$$

where $T$ is the disease-event time, and write $C$ for the available clinical history and $G$ for germline genomic state. Let $\mathbb{P}_1(\cdot) = \mathbb{P}(\cdot \mid Y = 1)$ and $\mathbb{P}_0(\cdot) = \mathbb{P}(\cdot \mid Y = 0)$. Assume throughout that the case distributions used below are absolutely continuous with respect to their control counterparts, so the relevant likelihood ratios exist as Radon–Nikodym derivatives and satisfy $\mathbb{E}_0[\Lambda] = 1$. Where the deterministic two-sided bounded-residual corollary is invoked, the corresponding conditional genomic laws are mutually absolutely continuous. For readability we write these derivatives using densities with respect to common dominating measures.

Define the clinical likelihood ratio

$$\Lambda_C(c) = \frac{p(c \mid Y = 1)}{p(c \mid Y = 0)}, \tag{2}$$

and the residual genomic likelihood ratio conditional on clinical history

$$\Lambda_{G|C}(g; c) = \frac{p(g \mid Y = 1, C = c)}{p(g \mid Y = 0, C = c)}. \tag{3}$$

The full joint likelihood ratio is

$$\Lambda_{C,G}(c, g) = \frac{p(c, g \mid Y = 1)}{p(c, g \mid Y = 0)}. \tag{4}$$

By the probability chain rule,

$$\boxed{\Lambda_{C,G}(c, g) = \Lambda_C(c)\, \Lambda_{G|C}(g; c).} \tag{5}$$

No conditional-independence assumption is required. The quantity $\Lambda_{G|C}$ is therefore the disease evidence supplied by genomics that remains after conditioning on the clinical channel. Likelihood-ratio thresholding is the classical optimal test for simple hypotheses in the Neyman–Pearson framework[14]; likelihood ratios also have a standard clinical interpretation as multiplicative updates of disease odds[15].

## III. Latent-state interpretation and calibrated-score special case

The likelihood-ratio formulation above is the primary statistical representation used in the theorem. A latent-state view provides biological interpretation but is not required for the result. Let $B_t$ denote the latent biological state at time $t$, $G$ the germline genomic state, and $E_{0:t}$ the accumulated non-genetic exposure history. The biological state evolves under the influence of $G$ and $E_{0:t}$. Define the disease-risk functional

$$\rho_h(B_t) = \mathbb{P}(T \le t + h \mid T > t, B_t). \tag{6}$$

The clinical event sequence $C_{0:t}$ is a noisy observation of the latent biological trajectory $B_{0:t}$. In an idealized Bayesian limit, a calibrated clinical-history predictor would satisfy

$$Z_h(C_{0:t}) = \mathbb{P}(T \le t{+}h \mid C_{0:t}) = \int \rho_h(b)\, d\mathbb{P}(B_t = b \mid C_{0:t}). \tag{7}$$

This idealized identity is not assumed for the raw Colorado ZeBRA score. Empirically, ZeBRA is treated as an ordinal clinical-history score and mapped directly to a score-level likelihood ratio.

If a clinical score $Z$ *is* calibrated so that $Z = \mathbb{P}(Y = 1 \mid C)$, and $\pi = \mathbb{P}(Y = 1)$, Bayes' rule yields

$$\frac{Z}{1 - Z} = \frac{\pi}{1 - \pi} \Lambda_C(C), \tag{8}$$

so thresholding a calibrated posterior is equivalent to thresholding $\Lambda_C$. Calibration is distinct from discrimination and is essential if a raw model score is to be interpreted as an absolute probability[16]. This calibrated-posterior mapping is therefore used only as a special theoretical corollary below; it is not the primary empirical route in this paper.

## IV. Main theorem

The core result is finite-stage boundary localization. At any fixed observed clinical-history stage, the relevant quantity is the high-probability scale of the residual genomic log-likelihood ratio at that stage.

*Notation.:* For sequential clinical histories $C_n = (X_1, \ldots, X_n)$, write

$$R_{G,n} = \log \Lambda_{G|C_n}(G; C_n) \tag{9}$$

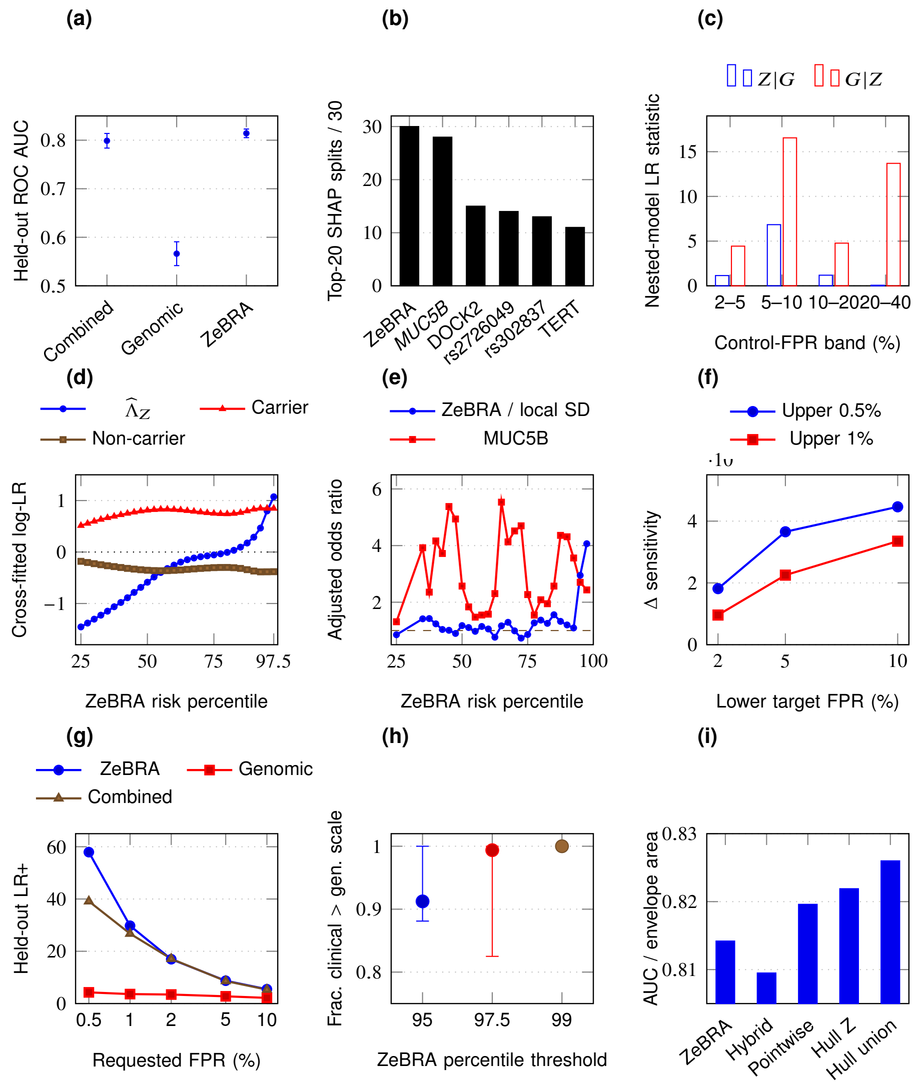


Fig. 1: **Integrated FILD/FILA evidence across global, local, likelihood-ratio, and operational regimes.** (a) Held-out discrimination shows that ZeBRA is substantially stronger globally than the broader array-derived genomic panel and that the nonlinear combined learner does not improve on ZeBRA. (b) Repeated-split SHAP attribution nevertheless shows that the combined learner uses the genomic channel: ZeBRA is the dominant feature and rs35705950 *MUC5B* is the most recurrent genomic contributor. (c) Directional nested-model likelihood-ratio/deviance statistics demonstrate localized residual *MUC5B* information after conditioning on ZeBRA. (d) The actual cross-fitted score-level evidence channels show clinical log-LR rising strongly with ZeBRA state while carrier and non-carrier residual *MUC5B* evidence remains state dependent and on a smaller scale. (e) Local adjusted effect sizes show persistent *MUC5B* contribution through much of the score axis and increasing clinical contribution in the extreme tail. (f) Boundary-targeted *MUC5B* rescue produces positive sensitivity gain over a training-FPR-matched ZeBRA comparator. (g) At stringent requested FPRs, ZeBRA has substantially larger held-out LR+ than the genomic-only learner. (h) Across the nine prespecified LR estimators, clinical log-LR exceeds the measured residual genomic scale for most subjects in the observed upper ZeBRA tail, approaching complete dominance by the 99th percentile. (i) Supporting ROC-envelope quantities show modest geometric complementarity among observed operating points; hull/envelope areas are descriptive and are not performance estimates from a separately validated classifier.

for the residual genomic log-likelihood ratio at stage $n$. For a clinical likelihood-ratio threshold $\lambda > 0$, define

$$D_{C_n} = \mathbf{1}\{\Lambda_{C_n} \geq \lambda\}, \qquad D_{C_n,G} = \mathbf{1}\{\Lambda_{C_n}\Lambda_{G|C_n} \geq \lambda\}. \tag{10}$$

For each fixed $n$ and tolerance $\delta \in (0, 1)$, define the finite-stage residual-evidence scale

$$b_{\delta,n} = \inf\left\{b \geq 0 : \mathbb{P}_0\big(|R_{G,n}| > b\big) \leq \delta\right\}, \tag{11}$$

whenever this quantile is finite. Thus $b_{\delta,n}$ is a property of the residual genomic evidence distribution at the clinical stage actually being analyzed.

**Theorem 1** (Finite-stage boundary localization). *Assume the relevant likelihood ratios are well defined and $b_{\delta,n} < \infty$ at the finite stage under consideration.*

(i) Exact factorization. *For every $n$,*

$$\Lambda_{C_n,G}(c_n, g) = \Lambda_{C_n}(c_n)\Lambda_{G|C_n}(g; c_n). \tag{12}$$

(ii) High-probability boundary localization. *For every $n$, every decision threshold $\lambda > 0$, and every $\delta \in (0, 1)$,*

$$\boxed{\mathbb{P}_0\Big(D_{C_n,G} \neq D_{C_n},\ \big|\log \Lambda_{C_n} - \log \lambda\big| > b_{\delta,n}\Big) \leq \delta.} \tag{13}$$

*Thus, at a fixed clinical stage, genomic information can reverse the clinical decision outside a log-LR neighborhood of half-width $b_{\delta,n}$ only on a control-side exceptional event of probability at most $\delta$.*

*Proof.* Part (i) is the chain rule:

$$\begin{aligned} \Lambda_{C_n,G}(c_n, g) &= \frac{p(c_n, g \mid Y = 1)}{p(c_n, g \mid Y = 0)} && (14)\\ &= \frac{p(c_n \mid Y = 1)p(g \mid Y = 1, c_n)}{p(c_n \mid Y = 0)p(g \mid Y = 0, c_n)} && (15)\\ &= \Lambda_{C_n}(c_n)\Lambda_{G|C_n}(g; c_n). && (16) \end{aligned}$$

For part (ii), if both $|\log \Lambda_{C_n} - \log \lambda| > b_{\delta,n}$ and $|R_{G,n}| \leq b_{\delta,n}$ hold, then adding $R_{G,n}$ cannot change the sign of $\log \Lambda_{C_n} - \log \lambda$. Hence

$$\Big\{D_{C_n,G} \neq D_{C_n},\ |\log \Lambda_{C_n} - \log \lambda| > b_{\delta,n}\Big\} \subseteq \{|R_{G,n}| > b_{\delta,n}\}, \tag{17}$$

whose probability is at most $\delta$ by the definition of $b_{\delta,n}$. □

## A. Finite-stage corollaries

**Corollary 1** (Deterministic bounded-residual special case). *Fix a clinical representation $C$. If there exist constants $0 < m \leq 1 \leq M < \infty$ such that*

$$0 < m \leq \Lambda_{G|C}(g; c) \leq M < \infty \tag{18}$$

*for $\mathbb{P}_0$-almost every relevant $(g, c)$, then genomic information can change which side of threshold $\lambda$ an observation occupies only if*

$$\boxed{\frac{\lambda}{M} \leq \Lambda_C(c) \leq \frac{\lambda}{m}.} \tag{19}$$

*Equivalently,*

$$\boxed{-\log M \leq \log \Lambda_C(c) - \log \lambda \leq -\log m.} \tag{20}$$

*The deterministic decision-relevant band has fixed width $\log(M/m)$ on the clinical log-likelihood scale.*

The proof is given in Supplementary Material, Section S1.

**Corollary 2** (Posterior-risk form for a calibrated clinical score). *If a clinical score $Z$ is calibrated so that $Z = \mathbb{P}(Y = 1 \mid C)$, let $\tau \in (0, 1)$ be any decision threshold and define posterior-odds threshold $\kappa = \tau/(1 - \tau)$. Under Corollary 1, genomic information can change the decision only if*

$$\boxed{\frac{\kappa}{M} \leq \frac{Z}{1 - Z} \leq \frac{\kappa}{m}.} \tag{21}$$

*Equivalently,*

$$-\log M \leq \operatorname{logit}(Z) - \operatorname{logit}(\tau) \leq -\log m. \tag{22}$$

*This is an idealized calibrated-score corollary, not the empirical representation used in the Colorado analysis. The raw ZeBRA output is not calibrated as a posterior probability there; the empirical analysis instead estimates a score-level likelihood ratio directly.*

The proof is given in Supplementary Material, Section S1.

## B. Local decision gain and global discrimination

**Proposition 1** (Local decision gain can coexist with negligible global AUC change). *Let $S(C)$ be a clinical score and $S'(C, G)$ an augmented score. Suppose $S' = S$ except on a measurable region $B$ of the clinical–genomic state space. Then*

$$\begin{aligned} |\mathrm{AUC}(S') - \mathrm{AUC}(S)| &\leq \mathbb{P}_1(B) + \mathbb{P}_0(B) - \mathbb{P}_1(B)\mathbb{P}_0(B) && (23)\\ &\leq \mathbb{P}_1(B) + \mathbb{P}_0(B). && (24) \end{aligned}$$

*Hence an augmented rule can produce a substantial change in decisions within a small boundary region while producing an arbitrarily small change in global AUC as the case and control mass of that region tends to zero.*

The proof is given in Supplementary Material, Section S1.

# V. Consequences and Scope of the Theory

Theorem 1 is the central result of the paper. It is finite-stage, requires only a finite control-side residual-genomic quantile $b_{\delta,n}$ at the clinical stage being analyzed, and is the part directly instantiated by the Colorado data.

## A. Clinical evidence accumulates sequentially

For a sequential history $C_n = (X_1, \ldots, X_n)$, the clinical likelihood ratio factorizes exactly as

$$\Lambda_C(C_n) = \prod_{k=1}^{n} \frac{p(X_k \mid Y = 1, X_{<k})}{p(X_k \mid Y = 0, X_{<k})}, \tag{25}$$

so that

$$\log \Lambda_C(C_n) = \sum_{k=1}^{n} \log \frac{p(X_k \mid Y=1, X_{<k})}{p(X_k \mid Y=0, X_{<k})}. \tag{26}$$

This is the standard likelihood-ratio accumulation underlying sequential testing and information accumulation[17,18]. It shows how additional clinical observations can add evidence. The manuscript makes no asymptotic claim that clinical log-likelihood ratios become mathematically unbounded as history lengthens; the sequential factorization is motivation for longitudinal evidence accumulation, not an additional assumption of Theorem 1.

### *B. Local decision gain need not produce global discrimination gain*

Theorem 1(ii) controls where genomics can change a threshold decision, not the global ranking of all case–control pairs. If genomic information matters only in a narrow decision band, a clinically meaningful subset of decisions can change while global AUC changes negligibly. Proposition 1 formalizes this distinction. This is closely related to the broader biomarker literature showing that strong association need not imply strong discrimination and that incremental AUC can be insensitive when an existing model already performs well[11–13]. The relevant empirical question is therefore not whether a particular combined learner outperforms ZeBRA globally, but whether genomics contributes reproducible residual evidence after conditioning on the clinical channel and whether that evidence is localized near clinically relevant boundaries.

Weak ZeBRA–genotype association is also not evidence that residual genomic disease information is absent. The relevant object is $\Lambda_{G|C}$, not the marginal dependence between a clinical score and genotype. ZeBRA can be nearly uninformative about rs35705950 carrier status while *MUC5B* still changes disease odds conditional on a given clinical state.

### *C. Operating thresholds and calibrated posterior scores are special cases*

The decision-relevant genomic region is defined relative to a chosen likelihood-ratio threshold. There is therefore no threshold-free universal "useful band." Corollary 2 translates this geometry into posterior-risk space only when a clinical score is calibrated so that $Z = \mathbb{P}(Y = 1 \mid C)$. Because raw Colorado ZeBRA is not calibrated as an absolute posterior probability, that corollary is not applied numerically. The empirical analysis instead estimates $\Lambda_Z$ directly from the observed ZeBRA score. This separation between discrimination and probability calibration follows standard prediction-model evaluation practice[16].

## VI. Empirical correspondence in the Colorado FILD/FILA analysis

### *A. Cohort and analysis hierarchy*

The empirical study uses a Colorado genomic–clinical cohort with linked longitudinal ZeBRA scores and a processed germline genotype matrix. Two genomic representations are kept conceptually distinct. For the global genomic-only and nonlinear combined-model analyses, the broader array-derived genomic feature panel is the processed matrix historically constructed by matching a prior FILD/FILA biological-driver SNP list and an additional candidate-locus list to the Colorado genotype array, then one-hot encoding the observed genotype states. The supplemental list contained 3,113 candidate rows, of which 148 additional raw genotype columns matched the array before union with the earlier disease-driver set. For the mechanistic likelihood-ratio, local-information, reconstruction, and rescue analyses, the genomic channel is instead the single focal *MUC5B* variant rs35705950 represented by T-carrier status. Supplementary Table S1 gives the construction, encoding, and analysis role of each representation, and the complete 171-locus retained panel is listed in Supplementary Table S10. This manuscript analyzes a single operational endpoint: future fibrotic interstitial lung disease or fibrotic interstitial lung abnormality (FILD/FILA). Subjects without a usable 104-week ZeBRA score are excluded. Controls are subjects with no FILD/FILA-defining fibrotic codes present in the operational phenotype, and false-positive rates use this full code-negative control group. A narrower subset with an explicit negative adjudication entry is retained only as a secondary sensitivity analysis.

The common repeated-split cohort contains 12,825 subjects, including 254 FILD/FILA-positive subjects. The direct *MUC5B* likelihood-ratio analysis further requires a called rs35705950 genotype, yielding 12,766 subjects and 252 FILD/FILA-positive subjects. The present paper deliberately does not analyze the sparse nonfibrotic endpoints explored during development; restricting the empirical argument to FILD/FILA keeps the phenotype aligned with the biological role of *MUC5B* and with the likelihood-ratio analysis used to test the theory.

The empirical analyses follow the theoretical hierarchy. Repeated held-out splits characterize global discrimination, while repeated-split SHAP attribution establishes which channels the nonlinear combined learner actually uses. Local analyses map where residual *MUC5B* predictive contribution is concentrated along the ZeBRA axis. The primary direct analysis then estimates the observable score-level likelihood-ratio channels, and the operating analyses ask whether localized genomic information can alter selected decisions. These complementary views are deliberately integrated into a single nine-panel FILD/FILA figure (Fig. 1). The explicit ZeBRA-to-*MUC5B* reconstruction check, residual-scale robustness, decision-flip localization, rescue FPR, stringent sensitivity, and adjudication-documented code-negative sensitivity analysis are consolidated in Supplementary Fig. S1; numerical details remain in tables where they are more legible. Table I summarizes how each theoretical component maps to the corresponding empirical analysis and its intended interpretation.

### *B. Global discrimination, incremental value, and feature attribution*

Across 30 repeated held-out outer splits, mean FILD/FILA AUC is $0.8142 \pm 0.0089$ for ZeBRA, $0.5662 \pm 0.0245$ for the broader array-derived genomic panel, and $0.7988 \pm 0.0150$ for the nonlinear combined learner (Fig. 1a). Thus the evaluated genomic panel is a much weaker global predictor than the clinical-history score, and the nonlinear combined learner does not improve on ZeBRA. The latter is a model-specific observation rather than a requirement of the theory.

The cleaner global test is the paired regularized analysis, in which ZeBRA-only and ZeBRA-plus-genomics models are evaluated on the same held-out subjects. The mean paired FILD/FILA AUC difference is only $+0.00043$, the median is 0, and the split-wise 2.5th–97.5th percentile range is $[-0.00507, 0.00408]$. Average precision decreases by 0.0311 on average, while Brier score and log loss worsen by $3.12 \times 10^{-4}$ and 0.00468, respectively. These are empirical split distributions rather than confidence limits from independent samples. The result is therefore not that the genomic variables are absent from the fitted models, but that adding the broader array-derived panel does not produce reproducible global incremental discrimination beyond ZeBRA in this cohort.

The repeated-split SHAP analysis makes that distinction explicit (Fig. 1b). In the combined FILD/FILA learner, the ZeBRA risk score is the top-ranked feature in every outer split and appears in the SHAP top 20 in 30/30 splits. The rs35705950 GG indicator is the most recurrent genomic feature, appearing in the top 20 in 28/30 splits and having mean rank 4.46 when present. By comparison, the next most recurrent displayed genomic features appear in only 15, 14, 13, and 11 of 30 splits. Thus the fitted combined learner has a stable hierarchy: the clinical-history score dominates feature attribution, while the *MUC5B* locus is the leading reproducible genomic contributor. SHAP attribution is a model-specific measure of predictive contribution rather than causal importance, but it rules out the simple explanation that the absence of global AUC gain arose because the combined learner failed to use the major genomic signal. The complete repeated-split recurrence summary is given in Supplementary Table S2.

### *C. ZeBRA does not reconstruct MUC5B*

Among 12,766 subjects with called rs35705950 genotype, ZeBRA predicts T-carrier status at essentially chance level: pooled AUC 0.5088, adjudication-documented code-negative AUC 0.4847, and FILD/FILA-positive AUC 0.4817 (Supplementary Fig. S1a). Carrier prevalence is nevertheless 39.3% in FILD/FILA-positive subjects versus 19.2% in the adjudication-documented code-negative subset. The pooled top 1% ZeBRA tail shows 1.47-fold carrier enrichment, but the corresponding enrichment is absent within disease strata. This pattern is consistent with shared association to disease status rather than recovery of genotype from clinical history. Together with the SHAP result, it establishes two distinct facts: the combined learner can use *MUC5B*, but the ZeBRA score itself does not encode the genotype.

### *D. Residual genomic information is localized in ZeBRA score space*

The absence of global incremental AUC does not imply that the genomic channel is conditionally uninformative everywhere. We therefore examined clinically interpretable control-FPR bands and overlapping moving windows along the ZeBRA percentile axis. Here the reported "nested-model LR statistic" is the likelihood-ratio/deviance statistic for adding one predictor after the other within a local subset. It is evidence for conditional predictive contribution, not an estimate of the individual-level likelihood-ratio factor $\Lambda_{G|C}$ in Theorem 1. The direct score-level LR analysis in the next subsection provides that closer empirical analogue.

The FPR-defined analysis makes the directional asymmetry explicit (Fig. 1c). In every displayed band, the nested-model statistic for adding *MUC5B* after ZeBRA ($G \mid Z$) exceeds the converse statistic for adding ZeBRA after *MUC5B* ($Z \mid G$). The difference is especially large in the 5–10% band, where $G \mid Z = 16.55$ versus $Z \mid G = 6.84$, and in the 20–40% band, where $G \mid Z = 13.69$ versus $Z \mid G = 0.064$ (Table II). The four non-overlapping $G \mid Z$ band tests are treated as one exploratory family and corrected by the Benjamini–Hochberg false-discovery-rate procedure; all four remain below $q = 0.05$ (Supplementary Table S4). The manuscript-wide rationale for which analyses do and do not constitute multiple-testing families is summarized in Supplementary Table S9. The overlapping moving-window summaries are not treated as separate formal hypothesis tests.

TABLE I: Theory–evidence correspondence for the FILD/FILA analysis. The empirical analyses test finite-stage consequences and operating behavior; they do not imply a universal genome-wide bound.

| Theory component | Empirical analysis | Empirical meaning |
|---|---|---|
| Distinct clinical and genomic channels | Repeated-split SHAP attribution and ZeBRA-to-*MUC5B* stratified analyses | The combined learner relies primarily on ZeBRA while retaining the *MUC5B* locus as the most stable genomic feature; weak global genomic increment is not explained by the learner ignoring *MUC5B* or by ZeBRA reconstructing genotype. |
| Finite-stage boundary localization (Theorem 1(ii)) | Cross-fitted $\widehat{\Lambda}_Z$, $\widehat{\Lambda}_{G\mid Z}$, $\widehat{b}_\delta$, local information curves, and decision-flip localization | The direct LR analysis estimates a control-side residual-genomic scale at the observed ZeBRA-score stage and tests whether genomic-induced decision changes remain boundary localized. |
| Local gain with little global AUC change (Proposition 1) | Paired incremental analysis, local conditional information, and matched-FPR rescue | Broad genomic augmentation yields essentially no global FILD/FILA AUC gain even though residual *MUC5B* information is substantial in selected score regimes and can alter selected decisions. |
| Observed high-score evidence dominance | Cross-fitted clinical-tail LR summaries and stringent held-out operating points | Over the observed extreme ZeBRA tail, clinical evidence exceeds the measured residual genomic scale. This is a finite-range empirical comparison, not an asymptotic claim. |
| Scope caveat | Same direct LR analyses | The stochastic bounds are estimated under the control law and for the measured *MUC5B* residual channel; they are not evidence of a universal genome-wide bound. |

The overlapping moving-window summaries show the same regime transition. Across much of the intermediate-to-high ZeBRA axis, residual *MUC5B* information exceeds residual ZeBRA information. At the 70th-percentile window, $G \mid Z = 10.93$ while $Z \mid G \approx 0.002$; at the 90th percentile the corresponding values are 19.78 and 1.23; and at 92.5 they are 20.37 and 0.41. The ordering reverses in the extreme clinical-risk tail: at the 95th-percentile window, $Z \mid G = 58.76$ versus $G \mid Z = 22.86$, and at 97.5 the values are 68.12 versus 15.61. Selected moving-window values are retained in Supplementary Table S5.

Effect-size summaries provide a complementary view in main Fig. 1e. FILD/FILA prevalence is consistently higher among *MUC5B* T-carriers than non-carriers within the moving windows. At the 70th-percentile window, prevalence is 4.28% in carriers versus 0.98% in non-carriers; at 90 it is 8.27% versus 2.05%; at 95 it is 18.45% versus 7.75%; and at 97.5 it is 20.48% versus 9.89%. The local *MUC5B* carrier OR remains several-fold across much of the score axis, whereas the ZeBRA OR per local SD rises sharply only in the extreme tail, reaching 2.95 at the 95th percentile and 4.07 at 97.5.

Within the narrow 5–10% control-FPR band, the observed ZeBRA AUC is 0.367. Conditional on membership in this restricted score interval, a randomly selected FILD/FILA case is therefore more likely than not to have a lower ZeBRA score than a randomly selected control—a local reversal of case–control rank ordering within the slice. This does not imply that ZeBRA is globally anti-predictive; global monotonic ranking need not persist after conditioning on a narrow portion of the score distribution. Because this band contains only 26 cases, we treat the reversal as a descriptive local feature. Full moving-window values and the multiplicity-adjusted local diagnostic tests are provided in the Supplementary Material.

### *E. Primary empirical test: direct score-level likelihood-ratio analysis*

The local moving-window statistics above describe conditional predictive contribution but are not themselves estimates of the individual-level likelihood-ratio factor in the theorem. The primary empirical test therefore instantiates the likelihood-ratio factorization directly at the observed ZeBRA-score level. Let $Z$ denote the ZeBRA score and $G \in \{0, 1\}$ indicate *MUC5B* T-carrier status. Then, without any sufficiency assumption,

$$\Lambda_{Z,G}(z, g) = \Lambda_Z(z)\Lambda_{G\mid Z}(g; z), \tag{27}$$

where

$$\Lambda_Z(z) = \frac{p(z \mid Y = 1)}{p(z \mid Y = 0)}, \tag{28}$$

$$\Lambda_{G\mid Z}(g; z) = \frac{p(g \mid Y = 1, Z = z)}{p(g \mid Y = 0, Z = z)}. \tag{29}$$

For carrier status, defining $q_y(z) = \mathbb{P}(G = 1 \mid Y = y, Z = z)$ gives

$$\Lambda_{G\mid Z}(1; z) = \frac{q_1(z)}{q_0(z)}, \qquad \Lambda_{G\mid Z}(0; z) = \frac{1 - q_1(z)}{1 - q_0(z)}. \tag{30}$$

These quantities are estimated out of fold by five repeats of fivefold cross-fitting with cubic-spline logistic models. The primary specification uses five spline knots and logistic regularization $C = 1$; a prespecified

TABLE II: Local FILD/FILA information in ZeBRA-defined control-FPR bands. Both directional nested-model LR/deviance statistics are shown so the asymmetry between channels is explicit. These local summaries are exploratory; the four formal $G \mid Z$ band tests are Benjamini–Hochberg FDR-adjusted as a single family in Supplementary Table S4.

| Band | $N$ | Cases | LR stat. $Z \mid G$ | LR stat. $G \mid Z$ | $G-Z$ partial | AUC ZeBRA | AUC *MUC5B* |
|---|---|---|---|---|---|---|---|
| 2–5% | 401 | 26 | 1.15 | 4.44 | 3.29 | 0.577 | 0.595 |
| 5–10% | 652 | 26 | 6.84 | 16.55 | 9.71 | 0.367 | 0.677 |
| 10–20% | 1,281 | 30 | 1.19 | 4.77 | 3.58 | 0.558 | 0.586 |
| 20–40% | 2,540 | 37 | 0.06 | 13.69 | 13.63 | 0.516 | 0.635 |

sensitivity grid crosses four, five, or six knots with $C \in \{0.3, 1, 3\}$.

Figure 1d shows the cross-fitted evidence channels themselves. The clinical score-level log likelihood ratio $\log \widehat{\Lambda}_Z$ increases monotonically over most of the ZeBRA axis and rises sharply in the extreme score tail. In contrast, the residual *MUC5B* channel is state dependent and remains on a much smaller scale: T-carrier status supplies positive residual evidence over most of the observed score axis, while non-carrier status supplies negative residual evidence. For example, in the 90th-percentile moving window the median clinical log-LR is 0.288, while the median carrier and non-carrier residual genomic log-LRs are 0.830 and $-0.369$; by the 97.5th-percentile window the median clinical log-LR is 1.073 while the corresponding genomic values remain approximately 0.847 and $-0.382$. Thus the cross-fitted curves directly visualize the additive log-LR geometry underlying the theorem: clinical evidence changes strongly with the clinical state, whereas the residual genomic perturbation remains comparatively constrained over the observed range.

For each control subject, define the statewise residual-genomic scale

$$B(Z) = \max_{g \in \{0,1\}} \left| \log \widehat{\Lambda}_{G|Z}(g; Z) \right|, \tag{31}$$

and let the empirical stochastic bound be the $(1-\delta)$ control quantile

$$\widehat{b}_\delta = Q_{1-\delta}[B(Z) \mid Y = 0]. \tag{32}$$

This is a score-level empirical analogue of the finite-stage scale $b_{\delta,n}$ in Theorem 1(ii). We write $\widehat{b}_\delta$ without an $n$ subscript to distinguish the observed score-level estimate from the theoretical $b_{\delta,n}$ defined conditional on the full clinical history $C_n$. The factorization $\Lambda_{Z,G} = \Lambda_Z \Lambda_{G|Z}$ is exact for the observable channel $Z$ and requires no sufficiency assumption. If $Z$ retained all disease-relevant information in $C_n$ needed for the residual genomic law, then $\Lambda_{G|Z}$ would coincide with $\Lambda_{G|C_n}$; we neither assume nor establish that stronger property.

In the primary 5-knot/$C$=1 specification, the control-side high-probability residual-genomic scales are $\widehat{b}_{0.05} = 0.8699$, $\widehat{b}_{0.01} = 0.9099$, and $\widehat{b}_{0.005} = 0.9241$ on the log-LR scale. Across all nine prespecified specifications, the corresponding ranges are 0.8441–0.9083, 0.8635–0.9617, and 0.8697–1.6100 (Supplementary Fig. S1b). The 95% and 99% scales are therefore stable, whereas the 99.5% tail is sensitive to the most flexible estimators. This pattern supports the high-probability finite-stage formulation of Theorem 1(ii), rather than a deterministic residual bound.

The most direct decision-level implication is also observed. When clinical decisions are thresholded at requested FPRs from 0.5% to 10%, genomic-induced decision changes remain concentrated near the clinical likelihood-ratio boundary (Supplementary Fig. S1c). Across the nine estimator specifications, the median boundary distance of flipped subjects is 0.1345–0.3039 log-LR units at requested FPR 0.5% and 0.2292–0.4991 at requested FPR 1%. Thus the same analysis that estimates the residual genomic scale also shows that the decisions changed by genomics are localized where the theorem predicts they should be.

The observed upper ZeBRA tail provides a separate descriptive comparison of clinical and residual-genomic evidence over the measured range. In the primary estimator, median clinical log-LR among subjects above the 95th, 97.5th, and 99th ZeBRA percentiles is 1.5175, 2.7325, and 3.8345, while the corresponding 95th percentiles of the estimated statewise residual genomic scale are 0.9213, 0.7758, and 0.7384. The fractions for which clinical log-LR exceeds that genomic scale are 0.9045, 0.9938, and 1.0000, respectively. These are finite-range empirical observations, not claims about asymptotic behavior as clinical history lengthens. The estimator-sensitivity range for this comparison is shown in main Fig. 1h and the numerical values are retained in Supplementary Table S3.

### *F. Secondary operating consequence*

The two-threshold *MUC5B* rescue analysis is retained as an application of the localization result rather than as its primary empirical support. Upper and lower ZeBRA thresholds are learned from training negatives; *MUC5B* rescue is applied only to untouched outer-test subjects between those thresholds; and a ZeBRA-only threshold is separately selected on training data to match the rescue rule's training-set FPR. Across all six evaluated threshold pairs, mean held-out sensitivity gain over this matched-ZeBRA comparator is positive. For the 0.5%/10% rule, mean sensitivity increases by 0.0446 while mean held-out FPR differs from matched ZeBRA by only $-2.53\times10^{-4}$. The sensitivity pattern is shown in main Fig. 1f; the near-zero FPR changes are shown in Supplementary Fig. S1d and numerical values are retained in Supplementary Table S6. Because rescue introduces additional threshold choices, it is treated as a secondary operating illustration rather than primary evidence for the theorem. The smaller adjudication-documented code-negative subset is reported only as a sensitivity analysis and does not redefine the primary control group.

Stringent operating-point and calibration analyses are treated similarly. At requested FPR 0.5%, mean held-out LR+ is approximately 57.9 for ZeBRA, 4.28 for the genomic-only model, and 39.1 for the nonlinear combined model; at 1%, the corresponding means are 29.8, 3.62, and 26.7. LR+ across all five requested FPRs is shown in main Fig. 1g, while the corresponding held-out sensitivities are shown in Supplementary Fig. S1e. The raw ZeBRA score is also not calibrated as an absolute FILD/FILA probability in this cohort, so the empirical theory–data connection uses the directly estimated score-level likelihood ratio $\Lambda_Z$, not a posterior-probability interpretation of the raw score. Calibration remains tabulated in Supplementary Table S8. The ROC-hull diagnostic is retained as main Fig. 1i solely as a geometric summary of observed operating points and is not used as primary evidence for the theorem.

## VII. Scope and limitations

The theory is a statement about information channels, not about a particular machine-learning architecture. It does not require independence between genomics and clinical history; dependence is retained explicitly through $\Lambda_{G|C}$. The central theorem is finite-stage and requires a finite residual-genomic quantile at the clinical stage being analyzed.

Several qualifications are essential. First, the direct empirical likelihood-ratio analysis estimates a finite-stage control-side residual-genomic scale for the measured *MUC5B* channel. It does not establish the same bound under the case law and does not imply a universal genome-wide bound. Second, the deterministic bounded-residual corollary remains a stronger special case. Third, Theorem 1(ii) concerns threshold reversals, not all posterior-risk changes. Fourth, the current global genomic comparator is the broader array-derived panel defined in Supplementary Table S1, whereas the direct residual-LR analysis focuses on *MUC5B*; neither should be interpreted as an empirical bound on genome-wide predictors, external polygenic risk scores, or unmeasured genomic information. Fifth, the raw ZeBRA score is not calibrated as an absolute posterior probability in this cohort, so Corollary 2 is not numerically applied; the empirical connection instead uses direct score-level LR estimation. Sixth, the four non-overlapping local $G \mid Z$ band tests are exploratory and are controlled as one family using Benjamini–Hochberg FDR adjustment; the overlapping moving windows are descriptive localization summaries rather than a separate family of formal tests. Their nested-model LR/deviance statistics are evidence of conditional predictive contribution, not estimates of the individual-level residual likelihood ratio in the theorem. Seventh, FILD/FILA controls are defined by absence of the fibrotic codes used by the operational phenotype; the smaller adjudication-documented code-negative subset is retained only as a secondary sensitivity analysis and does not redefine the primary control group. Finally, all rescue and direct-LR analyses remain internal and require prospective or external validation.

A stronger future validation should calibrate ZeBRA explicitly onto a likelihood-ratio or posterior-probability scale, estimate uncertainty for the direct LR quantities in an external cohort, evaluate prespecified fixed-specificity thresholds prospectively, externally replicate the operating-point analyses, and test whether the same boundary-localized geometry appears for other disease/genomic pairs.

## VIII. Conclusion

The central result is not that clinical history "contains" genomics, nor that genomics is globally useless. The two modalities are distinct evidence channels. Bayes' rule factorizes their joint evidence into a clinical likelihood ratio and a residual genomic likelihood ratio conditional on clinical state. The main theorem shows that, at a fixed clinical stage, the high-probability scale of residual genomic evidence determines a neighborhood of the clinical decision boundary outside which genomic-induced decision reversals are rare. The result is explicitly finite-stage: it characterizes the observed clinical stage without making an asymptotic claim about arbitrarily long histories.

The Colorado FILD/FILA analysis directly instantiates that geometry at the observed ZeBRA-score level. Broad genomic augmentation produces essentially zero paired incremental AUC after conditioning on ZeBRA. Yet the combined learner is not ignoring genomic information: repeated-split SHAP attribution ranks the ZeBRA risk score first in every split, while the rs35705950 *MUC5B* genotype indicator is the most recurrent genomic feature. ZeBRA itself does not reconstruct *MUC5B*, residual *MUC5B* predictive contribution varies strongly across the ZeBRA axis, and the cross-fitted evidence curves show distinct clinical and residual-genomic likelihood-ratio channels. The direct LR analysis estimates stable 95% and 99% control-side residual-genomic scales and shows that genomic-induced decision flips remain close to the clinical boundary.

The observed extreme ZeBRA tail also shows clinical log-LR dominating the measured residual genomic scale, while boundary-targeted rescue provides a secondary operational illustration of how localized genomic information can improve selected decisions. The tail comparison is an empirical finite-range observation rather than an asymptotic theorem. The resulting picture is one of *regime-dependent complementarity*: genomics can be biologically and locally predictive even when its global incremental discrimination beyond a strong longitudinal clinical-history channel is minimal.

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## Supplementary Material

This Supplementary Material is part of the same document as the main manuscript. It contains auxiliary proofs and the additional empirical analyses needed to assess genomic-panel provenance, robustness, local diagnostics, operating behavior, and calibration. The empirical scope remains restricted to the FILD/FILA endpoint. The broader genomic feature panel used for global model comparisons is distinguished explicitly from the focal *MUC5B* rs35705950 channel used for the direct likelihood-ratio and local mechanistic analyses. The graphical results displaced from the nine-panel main figure are consolidated into Supplementary Fig. S1; complete numerical values remain in the accompanying tables where a figure would be less concise.

### *S1. Proofs of auxiliary results*

*Proof of Corollary 1:*

*Proof.* Under (18), $R_G = \log \Lambda_{G|C}$ lies in $[\log m, \log M]$ almost surely. If $\Lambda_C < \lambda/M$, even the largest genomic factor cannot cross the threshold; if $\Lambda_C > \lambda/m$, even the smallest genomic factor cannot move the observation below threshold. This gives (19)–(20). □

*Proof of Corollary 2:*

*Proof.* Equation (8) shows that posterior odds differ from $\Lambda_C$ only by the positive constant prior odds $\pi/(1-\pi)$. Substituting into (19) yields the result. □

*Proof of Proposition 1:*

*Proof.* AUC is the probability that a randomly drawn case receives a higher score than a randomly drawn control, with the usual half-credit convention for ties. If neither member of a case–control pair lies in $B$, then $S' = S$ for both members and the pairwise ranking contribution is unchanged. Therefore the AUC contribution can differ only when at least one member of the pair lies in $B$. The probability of that event is

$$1 - (1 - \mathbb{P}_1(B))(1 - \mathbb{P}_0(B)) = \mathbb{P}_1(B) + \mathbb{P}_0(B) - \mathbb{P}_1(B)\mathbb{P}_0(B), \quad (33)$$

which bounds the maximum possible AUC change. □

### *S2. Genomic panel construction and analysis roles*

The phrase "genomic panel" in the main text refers to the processed array-derived feature matrix used for the broad genomic-only and combined prediction analyses, not to the single-locus representation used in the direct likelihood-ratio theory test. Historically, the broader matrix was generated by matching a FILD/FILA disease-driver SNP list and an additional candidate-locus list to the Colorado genotype array and then one-hot encoding categorical genotype states. The retained disease-driver list contains 26 variants. The archived supplemental candidate list contained 3,113 rows; 148 additional raw genotype columns matched the array before union with the disease-driver set. The final processed participant-by-feature matrix is the retained analysis input used throughout the broad genomic analyses.

For rs35705950, the historical one-hot columns encode GG, GT, and TT states separately. A valid called genotype has exactly one genotype-state indicator equal to one; an all-zero row denotes unavailable/missing genotype and is not treated as TT or as a T-carrier. For the direct likelihood-ratio, local-information, reconstruction, and rescue analyses, this three-state encoding is reduced to the biologically interpretable T-carrier indicator (GT or TT versus GG) among subjects with a called genotype.

### *S3. Repeated-split SHAP stability*

The nonlinear combined learner was evaluated across 30 outer splits. SHAP values were computed within each fitted split and features were ranked by mean absolute SHAP magnitude. The main text reports recurrence in the top 20 because recurrence is more stable across refits than a single pooled magnitude. Table S2 shows the leading recurrent features for FILD/FILA. The ZeBRA risk score is rank 1 in every split. The rs35705950 GG indicator is the most recurrent genomic feature and appears in the top 20 in 28 of 30 splits. The result supports a hierarchy in which the clinical-history score dominates the combined learner while the *MUC5B* locus remains its most stable genomic contributor.

### *S4. Direct likelihood-ratio estimation and robustness*

For the direct score-level analysis, $Z$ is the ZeBRA score and $G$ is binary *MUC5B* T-carrier status. The subject-level quantities $\Pr(Y = 1 \mid Z)$, $\Pr(G = 1 \mid Y = 1, Z)$, and $\Pr(G = 1 \mid Y = 0, Z)$ are estimated out of fold by five repeats of fivefold stratified cross-fitting. Each regression uses a cubic spline expansion of the monotone logit transform of the bounded ZeBRA score followed by logistic regression. The primary estimator uses five spline knots and $C = 1$. The sensitivity analysis crosses knot counts $\{4, 5, 6\}$ with $C \in \{0.3, 1, 3\}$; this grid is prespecified as an estimator-robustness analysis rather than used for model selection.

TABLE S1: Definition and provenance of the genomic representations used in the FILD/FILA manuscript. The 148 count refers to additional raw genotype columns matched from the supplemental candidate list before union with the disease-driver set and before one-hot expansion; it is not the total number of final model columns.

| Representation | Source / construction | Encoded analysis variables | Manuscript role |
|---|---|---|---|
| FILD/FILA disease-driver set | Retained 26-variant biological-driver list matched to Colorado array headers by rsID or genomic coordinate; matched columns form the base of the processed matrix. | Categorical genotype calls, subsequently one-hot encoded. | Base component of the broader genomic-only and combined-model feature set. |
| Supplemental candidate expansion | 3,113 archived candidate rows were matched to array headers after deduplication/availability matching; 148 additional raw genotype columns matched before union with the disease-driver set. | Matched categorical genotype columns, subsequently one-hot encoded. | Expands the broader panel used for global discrimination, paired incremental analyses, and SHAP attribution. |
| Final broader genomic panel | Union of the matched disease-driver and supplemental genotype columns, represented in the retained processed participant-by-feature matrix. ZeBRA risk is merged separately and phenotype labels are not genomic predictors. | One-hot genotype-state indicators; unavailable original genotypes remain all-zero across the corresponding state indicators. | Genomic-only model and non-linear/regularized ZeBRA-plus-genomics models. |
| Focal mechanistic genomic channel | *MUC5B* rs35705950. Historical states are GG, GT, and TT; analyses requiring a called genotype exclude all-zero unavailable states. | Binary T-carrier status: GT or TT versus GG. | ZeBRA-to-genotype reconstruction check, local conditional-contribution analyses, direct $\Lambda_{G\|Z}$ estimation, and boundary-targeted rescue. |

TABLE S2: Repeated-split SHAP stability in the combined FILD/FILA model. Mean rank is calculated only among splits in which the feature appears in the top 20.

| Feature | Top-20 splits / 30 | Mean rank when top 20 |
|---|---|---|
| ZeBRA predicted risk | 30 | 1.00 |
| rs35705950 GG (*MUC5B*) | 28 | 4.46 |
| rs13183751 G1 | 15 | 9.27 |
| rs2726049 A1 | 14 | 9.93 |
| rs302837 A1 | 13 | 8.54 |
| rs2736100 A1 (*TERT*) | 11 | 10.27 |

The score-level factorization

$$\Lambda_{Z,G}(z,g) = \Lambda_Z(z)\Lambda_{G|Z}(g;z) \tag{34}$$

is exact for the observed channels. The control-side residual scale is

$$B(Z) = \max_g |\log \widehat{\Lambda}_{G|Z}(g;Z)|, \qquad \widehat{b}_\delta = Q_{1-\delta}[B(Z) \mid Y=0] \tag{35}$$

The analysis contains 12,766 subjects with called rs35705950 genotype, including 252 FILD/FILA cases. In the primary specification, $\mathbb{E}_0[\widehat{\Lambda}_Z]$ is normalized to 1 and the observed control mean of the joint factor $\widehat{\Lambda}_Z\widehat{\Lambda}_{G|Z}$ is 1.0149.

The larger sensitivity of the most flexible estimators in very sparse tails is why the manuscript emphasizes stable 95% and 99% residual scales and treats the 99.5% estimate more cautiously. Supplementary Fig. S1b shows the residual-scale robustness, while main Fig. 1h shows the corresponding upper-tail dominance fraction across all nine estimators. The high-ZeBRA tail remains an observed finite-range phenomenon and is not interpreted as an asymptotic claim about arbitrarily long clinical histories. The nine estimator specifications constitute a prespecified robustness grid rather than nine hypothesis tests; no multiplicity correction is applied to their displayed ranges.

### *S5. Local-information diagnostics and multiplicity control*

The local nested-model LR/deviance statistics are tests of conditional contribution, not estimates of the subject-level residual likelihood ratio. The four non-overlapping control-FPR bands form the only family of formal local hypothesis tests reported in the manuscript. We therefore apply the Benjamini–Hochberg procedure across these four reported exploratory $G \mid Z$ tests. The nominal p-values are 0.0352, $4.74 \times 10^{-5}$, 0.0289, and $2.15 \times 10^{-4}$ for the 2–5%, 5–10%, 10–20%, and 20–40% bands, respectively; the corresponding BH-FDR q-values are 0.0352, $1.896 \times 10^{-4}$, 0.0352, and $4.30 \times 10^{-4}$. Thus all four remain below $q = 0.05$. These remain exploratory localization tests: FDR control addresses multiplicity within this reported family but does not convert the analyses into an independent confirmatory study. The overlapping moving-window trajectories are treated as descriptive effect/localization summaries and are

TABLE S3: Direct empirical likelihood-ratio summaries. The primary column uses the five-knot, $C = 1$ estimator; the range column reports the minimum and maximum across the nine prespecified specifications.

| Quantity | Primary value | Robustness range |
|---|---|---|
| $\widehat{b}_{0.05}$ | 0.8699 | 0.8441–0.9083 |
| $\widehat{b}_{0.01}$ | 0.9099 | 0.8635–0.9617 |
| $\widehat{b}_{0.005}$ | 0.9241 | 0.8697–1.6100 |
| Median $\log \Lambda_Z$ above 95th percentile | 1.5175 | 1.4383–1.6069 |
| 95th pct. genomic scale above 95th percentile | 0.9213 | 0.8155–5.6485 |
| Fraction clinical > genomic scale above 95th percentile | 0.9045 | 0.8811–1.0000 |
| Median $\log \Lambda_Z$ above 97.5th percentile | 2.7325 | 2.3756–2.8043 |
| 95th pct. genomic scale above 97.5th percentile | 0.7758 | 0.7601–8.4206 |
| Fraction clinical > genomic scale above 97.5th percentile | 0.9938 | 0.8250–1.0000 |
| Median $\log \Lambda_Z$ above 99th percentile | 3.8345 | 3.7850–3.9379 |
| 95th pct. genomic scale above 99th percentile | 0.7384 | 0.6849–0.7473 |
| Fraction clinical > genomic scale above 99th percentile | 1.0000 | 1.0000–1.0000 |

not assigned a separate family of p-values. No univariate SNP-level p-values from the broader genomic panel are used as manuscript evidence; the broader-panel results are held-out predictive-performance and feature-attribution analyses, so there is no locus-wise multiple-testing family to correct in those results.

Main Fig. 1e shows the corresponding moving-window adjusted effect-size trajectories.

### *S6. Boundary-targeted rescue and stringent operating points*

The rescue rule uses an upper ZeBRA threshold and a lower boundary threshold learned from training negatives. Subjects above the upper threshold are positive regardless of genotype; subjects in the interval between the lower and upper thresholds are rescued only if they are *MUC5B* T-carriers. For each outer split, a separate ZeBRA-only comparator threshold is selected using training data to match the rescue rule's training-set FPR. All comparisons are then evaluated on untouched test subjects. Main Fig. 1f shows the sensitivity gains; Supplementary Fig. S1d shows the corresponding operational-FPR differences.

The explicitly adjudicated-negative diagnostic applies the already learned matched thresholds unchanged to the adjudicated-negative subset of each test split. Because this subset contains only about 216 subjects per split on average, its repeated-split distribution is treated as a control diagnostic rather than definitive evidence of superiority. Supplementary Fig. S1f shows the rescue-minus-matched adjudicated-negative FPR; the complete numerical intervals remain in Table S7.

For the 0.5%/10% rule, rescue has lower adjudicated-negative FPR than matched ZeBRA in 81% of outer splits; the corresponding fraction is 78% for the 1%/10% rule. Five of the six rules have a negative mean rescue-minus-matched FPR, while the 1%/2% rule is essentially equal.

At stringent requested false-positive rates, the clinical-history score remains substantially stronger than the genomic-only learner. Mean held-out LR+ at requested FPR 0.5% is approximately 57.9 for ZeBRA, 4.28 for the genomic-only model, and 39.1 for the nonlinear combined model. At requested FPR 1%, the corresponding values are 29.8, 3.62, and 26.7. Main Fig. 1g shows LR+ across all five requested FPRs; Supplementary Fig. S1e shows the corresponding held-out sensitivities. The rescue summaries and stringent operating curves are repeated-split performance summaries rather than collections of hypothesis tests, so no additional p-value multiplicity correction is applied to them.

### *S7. Calibration diagnostics*

The raw ZeBRA score is not an absolute calibrated FILD/FILA probability in this cohort. Across repeated held-out splits, its mean calibration slope is 0.180 and expected calibration error (ECE) is 0.570. The corresponding values are 0.241 and 0.115 for the genomic model and 1.140 and 0.121 for the nonlinear combined model. These diagnostics motivate treating ZeBRA as an ordinal clinical-history score and estimating $\Lambda_Z$ directly rather than interpreting the raw score as posterior probability.

### *S8. Supporting ROC-hull diagnostic*

A descriptive ROC-hull analysis was used only to characterize geometric complementarity among observed operating points. The empirical/interpolated AUCs are 0.8142 for ZeBRA and 0.8095 for the hybrid decision rule. The convex hulls of the individual curves have AUCs 0.8219 and 0.8182, respectively; the pointwise maximum has AUC 0.8196; and the convex hull of the concatenated ZeBRA and hybrid ROC points has AUC 0.8260. These hull quantities are envelopes over evaluated operating points, not held-out performance estimates from a separately

TABLE S4: Multiplicity adjustment for the four non-overlapping local $G \mid Z$ tests. Benjamini–Hochberg q-values control false discovery rate across the four reported exploratory control-FPR bands.

| Control-FPR band | Nominal p-value | BH-FDR q-value |
|---|---|---|
| 2–5% | 0.0352 | 0.0352 |
| 5–10% | $4.74 \times 10^{-5}$ | $1.896 \times 10^{-4}$ |
| 10–20% | 0.0289 | 0.0352 |
| 20–40% | $2.15 \times 10^{-4}$ | $4.30 \times 10^{-4}$ |

TABLE S5: Selected overlapping moving-window values illustrating the information crossover. Values are descriptive localization summaries.

| ZeBRA percentile | LR stat. $Z \mid G$ | LR stat. $G \mid Z$ | OR ZeBRA/local SD | OR *MUC5B* | Prev. carrier | Prev. non-carrier |
|---|---|---|---|---|---|---|
| 65.0 | 0.45 | 13.36 | 1.17 | 5.53 | 0.0466 | 0.00865 |
| 70.0 | 0.002 | 10.93 | 0.99 | 4.52 | 0.0428 | 0.00980 |
| 90.0 | 1.23 | 19.78 | 1.20 | 4.31 | 0.0827 | 0.0205 |
| 92.5 | 0.41 | 20.37 | 1.09 | 3.56 | 0.1067 | 0.0323 |
| 95.0 | 58.76 | 22.86 | 2.95 | 2.70 | 0.1845 | 0.0775 |
| 97.5 | 68.12 | 15.61 | 4.07 | 2.43 | 0.2048 | 0.0989 |

TABLE S6: Matched-FPR rescue results across 100 repeated outer splits. $\Delta$ values are rescue minus matched ZeBRA.

| Rule | Mean $\Delta$ sensitivity | Mean $\Delta$FPR |
|---|---|---|
| 0.5% / 2% | +0.01815 | $+1.20 \times 10^{-5}$ |
| 0.5% / 5% | +0.03656 | $-4.00 \times 10^{-5}$ |
| 0.5% / 10% | +0.04464 | $-2.53 \times 10^{-4}$ |
| 1% / 2% | +0.00960 | $+6.66 \times 10^{-6}$ |
| 1% / 5% | +0.02252 | $-1.84 \times 10^{-4}$ |
| 1% / 10% | +0.03351 | $-1.78 \times 10^{-4}$ |

TABLE S7: Explicitly adjudicated-negative comparison. FPR values are proportions; $\Delta$ is rescue minus matched ZeBRA.

| Rule | Baseline FPR | Rescue FPR | Matched FPR | Mean $\Delta$ | 2.5–97.5% $\Delta$ |
|---|---|---|---|---|---|
| 0.5% / 2% | 0.1292 | 0.1421 | 0.1458 | -0.0037 | -0.0249–0.0094 |
| 0.5% / 5% | 0.1292 | 0.1628 | 0.1689 | -0.0060 | -0.0353–0.0225 |
| 0.5% / 10% | 0.1292 | 0.1958 | 0.2144 | -0.0185 | -0.0561–0.0194 |
| 1% / 2% | 0.1548 | 0.1635 | 0.1628 | +0.0006 | -0.0142–0.0138 |
| 1% / 5% | 0.1548 | 0.1842 | 0.1888 | -0.0046 | -0.0315–0.0194 |
| 1% / 10% | 0.1548 | 0.2172 | 0.2364 | -0.0192 | -0.0523–0.0188 |

TABLE S8: Calibration summary for FILD/FILA.

| Model | Calibration slope | ECE |
|---|---|---|
| ZeBRA | 0.180 | 0.570 |
| Genomic | 0.241 | 0.115 |
| Combined | 1.140 | 0.121 |

trained classifier. Main Fig. 1i retains the most interpretable summary quantities solely as a geometric diagnostic; they are not used as primary evidence for the theorem.

*S9. Scope of the empirical likelihood-ratio result*

The empirical residual-genomic quantiles are fixed-score quantities for the observed ZeBRA representation and the measured *MUC5B* channel. They correspond to the finite-stage construction in Theorem 1(ii). The observed high-ZeBRA likelihood-ratio tail is reported only over the measured range and is not interpreted as an asymptotic statement about arbitrarily long histories or unbounded clinical evidence. Neither the direct LR result nor the broader array-derived genomic-panel analysis should be interpreted as an empirical bound on all genome-wide information, external polygenic scores, or unmeasured genomic features.

*S10. Genomic-panel inventory and multiplicity rationale*

The processed genomic matrix used by the broad genomic-only and ZeBRA-plus-genomics analyses contains 513 one-hot genotype-state predictor columns corresponding to 171 retained variant loci. This count is obtained directly from the header of the retained processed genomic matrix: each retained locus is represented by exactly three columns with state suffixes `_0`, `_1`, and `_2`. The complete retained-locus inventory is rendered in Supplementary Table S10. It lists all 171 rsIDs in matrix order, the processed allele tag, and manuscript annotations where explicitly used.

The 171-locus count is the final modeled panel and should not be confused with the intermediate source-list counts. The retained FILD/FILA disease-driver list contains 26 variants, while matching the archived supplemental candidate list produced 148 additional raw genotype columns before union and deduplication. Because the source sets can overlap and the final representation is formed after union and one-hot expansion, these intermediate counts should not be added to infer the final panel size. The focal mechanistic channel remains *MUC5B* rs35705950, which is also present in the broader panel; its three-state representation is reduced to GT/TT T-carrier status versus GG only for the direct mechanistic analyses.

Supplementary Table S9 summarizes the multiplicity treatment for each empirical analysis and makes explicit the distinction between formal families of null-hypothesis tests and predictive, attribution, robustness, or descriptive analyses.

The reason no panel-wide multiple-testing correction is applied to the broad genomic analyses is therefore substantive rather than merely procedural. The manuscript does not conduct 171 univariate SNP association tests and then select loci according to nominal significance. Instead, the complete encoded panel enters supervised multivariable learners whose generalization is evaluated on held-out subjects. The relevant protection against overfitting in that setting is separation of training and test data together with regularization/model selection confined to training data; applying Bonferroni or FDR across the 171 input features would answer a different inferential question—simultaneous SNP-wise association—and would not constitute a correction of held-out predictive performance. Likewise, reporting several performance metrics does not create a multiple-testing family when those metrics are used as complementary descriptive estimands rather than as a set of thresholded significance tests.

The same distinction applies to SHAP recurrence, moving-window trajectories, rescue operating points, and the direct-LR robustness grid: none is used as a collection of nominal p-values from which favorable results are selected. By contrast, the four non-overlapping local $G \mid Z$ analyses do produce and interpret four p-values addressing the same conditional-contribution question in the same cohort, so they are treated as one exploratory family and corrected by Benjamini–Hochberg FDR. If future versions of the manuscript add SNP-by-SNP association p-values, formal p-values across the overlapping moving windows, or multiple separately interpreted genomic hypotheses, those analyses would require an explicitly defined multiplicity family and corresponding adjustment.

TABLE S9: Multiplicity-control rationale across the empirical analyses. Multiple-testing correction is required when a family of null hypotheses is evaluated and inferential claims depend on the resulting p-values. It is not a generic correction for the number of predictors, performance metrics, model refits, or descriptive robustness summaries.

| Analysis | Statistical role | Formal p-value family? | Multiplicity treatment |
|---|---|---|---|
| Broad genomic-only and ZeBRA-plus-genomics models | Held-out predictive performance (AUC, average precision, Brier score, log loss) for multivariable models | No | None. The 171 loci are predictors inside a fitted model, not 171 separately interpreted null-hypothesis tests. |
| Repeated-split SHAP recurrence | Feature-attribution and stability summary across refits | No | None. SHAP ranks/recurrence are descriptive model-attribution quantities, not locus-wise p-values. |
| Paired repeated-split incremental analysis | Distribution of held-out performance differences on matched test subjects | No | None. Split-wise values summarize predictive stability and are not treated as independent hypothesis-test replicates. |
| Overlapping ZeBRA moving windows | Descriptive localization/effect-size trajectories | No | None. No window-specific significance threshold is used for the manuscript claim. |
| Nine direct-LR estimator specifications | Prespecified estimator-sensitivity/robustness grid | No | None. The displayed min–max ranges are robustness summaries, not nine competing null tests. |
| Matched-FPR rescue and stringent operating points | Held-out sensitivity/FPR/LR+ operating characteristics | No | None. These are performance estimands at evaluated operating rules, not a collection of p-value-based claims. |
| Four non-overlapping local $G \mid Z$ bands | Exploratory tests of conditional genomic contribution after ZeBRA | Yes: four tests | Benjamini–Hochberg FDR correction across the four reported band tests; all remain below $q = 0.05$. |

TABLE S10: Exhaustive retained genomic-panel inventory used in the broad genomic-only and ZeBRA-plus-genomics analyses. The processed matrix contains 171 retained loci, each represented by three one-hot genotype-state columns (513 predictor columns total). Entries are shown in the exact matrix order; the allele column is the processed header allele tag. *MUC5B* rs35705950 is the focal mechanistic locus and is also part of the broader panel.

| No. | Variant | Allele | No. | Variant | Allele | No. | Variant | Allele |
|---|---|---|---|---|---|---|---|---|
| 1 | rs3131520 | C | 58 | rs6899532 | G | 115 | rs74760554 | C |
| 2 | rs55993474 | A | 59 | rs10484326 | T | 116 | rs12275748 | C |
| 3 | rs2375892 | T | 60 | rs2076295 | T | 117 | rs664034 | T |
| 4 | rs10881263 | G | 61 | rs3778337 | G | 118 | rs10842996 | T |
| 5 | rs11102362 | G | 62 | rs4960330 | T | 119 | rs10783497 | A |
| 6 | rs16837903 | G | 63 | rs148106529 | G | 120 | rs117427783 | T |
| 7 | rs11264306 | A | 64 | rs2071627 | C | 121 | rs76739551 | C |
| 8 | rs7547997 | G | 65 | rs3130157 | G | 122 | rs78288787 | C |
| 9 | rs699240 | T | 66 | rs56184452 | G | 123 | rs12422371 | C |
| 10 | rs114064385 | G | 67 | rs75714781 | G | 124 | rs325093 | T |
| 11 | rs55682827 | C | 68 | rs11154034 | T | 125 | rs923077 | A |
| 12 | rs17019820 | A | 69 | rs76931164 | T | 126 | rs6491201 | C |
| 13 | rs2577254 | T | 70 | rs79242376 | G | 127 | rs117750427 | A |
| 14 | rs12710674 | C | 71 | rs9397724 | G | 128 | rs17372089 | A |
| 15 | rs77998999 | A | 72 | rs35677769 | G | 129 | rs9577395 | C |
| 16 | rs6736148 | C | 73 | rs12665063 | A | 130 | rs1278769 | G |
| 17 | rs4672388 | T | 74 | rs6970647 | T | 131 | rs17616659 | G |
| 18 | rs1123573 | A | 75 | rs2726049 | A | 132 | rs34825924 | G |
| 19 | rs17007563 | C | 76 | rs34262641 | A | 133 | rs10518693 | C |
| 20 | rs12713927 | G | 77 | rs2353082 | C | 134 | rs2034650 | A |
| 21 | rs11680126 | C | 78 | rs4727443 | C | 135 | rs144948432 | A |
| 22 | rs1669778 | T | 79 | rs62505367 | T | 136 | rs1496725 | A |
| 23 | rs1488496 | C | 80 | rs3887876 | C | 137 | rs12928193 | G |
| 24 | rs56387623 | T | 81 | rs68170294 | G | 138 | rs9931202 | G |
| 25 | rs2361278 | T | 82 | rs74987898 | C | 139 | rs1273431 | A |
| 26 | rs79252367 | G | 83 | rs10976897 | G | 140 | rs71389435 | C |
| 27 | rs1479945 | C | 84 | rs62530710 | G | 141 | rs6499642 | C |
| 28 | rs1015097 | A | 85 | rs10869195 | A | 142 | rs12446021 | G |
| 29 | rs4680134 | C | 86 | rs56744128 | A | 143 | rs28703652 | G |
| 30 | rs76160568 | A | 87 | rs1923802 | T | 144 | rs169889 | A |
| 31 | rs10776482 | A | 88 | rs1851741 | G | 145 | rs3744761 | C |
| 32 | rs56111559 | G | 89 | rs7041597 | T | 146 | rs17690703 | C |
| 33 | rs146764446 | A | 90 | rs13299670 | C | 147 | rs1981997 | G |
| 34 | rs2609255 | T | 91 | rs61829240 | G | 148 | rs415430 | T |
| 35 | rs2609260 | T | 92 | rs1189320 | G | 149 | rs73997403 | C |
| 36 | rs756345 | G | 93 | rs55672765 | C | 150 | rs918174 | T |
| 37 | rs6532150 | G | 94 | rs34547036 | C | 151 | rs75291377 | G |
| 38 | rs144128391 | T | 95 | rs10748858 | T | 152 | rs11083093 | G |
| 39 | rs148877274 | T | 96 | rs2067832 | G | 153 | rs66884960 | C |
| 40 | rs145802912 | A | 97 | rs11191865 | G | 154 | rs76475316 | T |
| 41 | rs151036604 | T | 98 | rs4918072 | G | 155 | rs893889 | G |
| 42 | rs1422506 | G | 99 | rs10886437 | C | 156 | rs12610495 | A |
| 43 | rs34625679 | T | 100 | rs616768 | G | 157 | rs2109069 | G |
| 44 | rs7730878 | A | 101 | rs7942850 | T | 158 | rs708686 | C |
| 45 | rs7725218 | G | 102 | rs7934606 | C | 159 | rs62112917 | C |
| 46 | rs2736100* | A | 103 | rs4077759 | T | 160 | rs73008477 | C |
| 47 | rs72722911 | G | 104 | rs7952154 | C | 161 | rs10422762 | A |
| 48 | rs77144736 | C | 105 | rs34425120 | G | 162 | rs41482151 | C |
| 49 | rs62354543 | A | 106 | **rs35705950**† | G | 163 | rs12975986 | C |
| 50 | rs72755986 | C | 107 | rs2857476 | C | 164 | rs74490361 | T |
| 51 | rs72762052 | C | 108 | rs3829223 | C | 165 | rs57625989 | A |
| 52 | rs116575227 | A | 109 | rs3750920 | C | 166 | rs302837 | A |
| 53 | rs469942 | C | 110 | rs5743894 | T | 167 | rs601878 | C |
| 54 | rs2228972 | G | 111 | rs5743890 | T | 168 | rs4811028 | G |
| 55 | rs12658068 | G | 112 | rs7122936 | A | 169 | rs12626788 | C |
| 56 | rs1423029 | A | 113 | rs61867586 | C | 170 | rs75764565 | C |
| 57 | rs13183751 | G | 114 | rs73409602 | C | 171 | rs45619034 | G |

† *MUC5B* rs35705950; * *TERT* rs2736100. A complete machine-readable version of the inventory is provided with the supplementary materials. The retained disease-driver source contains 26 variants; the supplemental matching count is an intermediate pre-union quantity, so source-list counts do not sum directly to the 171-locus final panel.

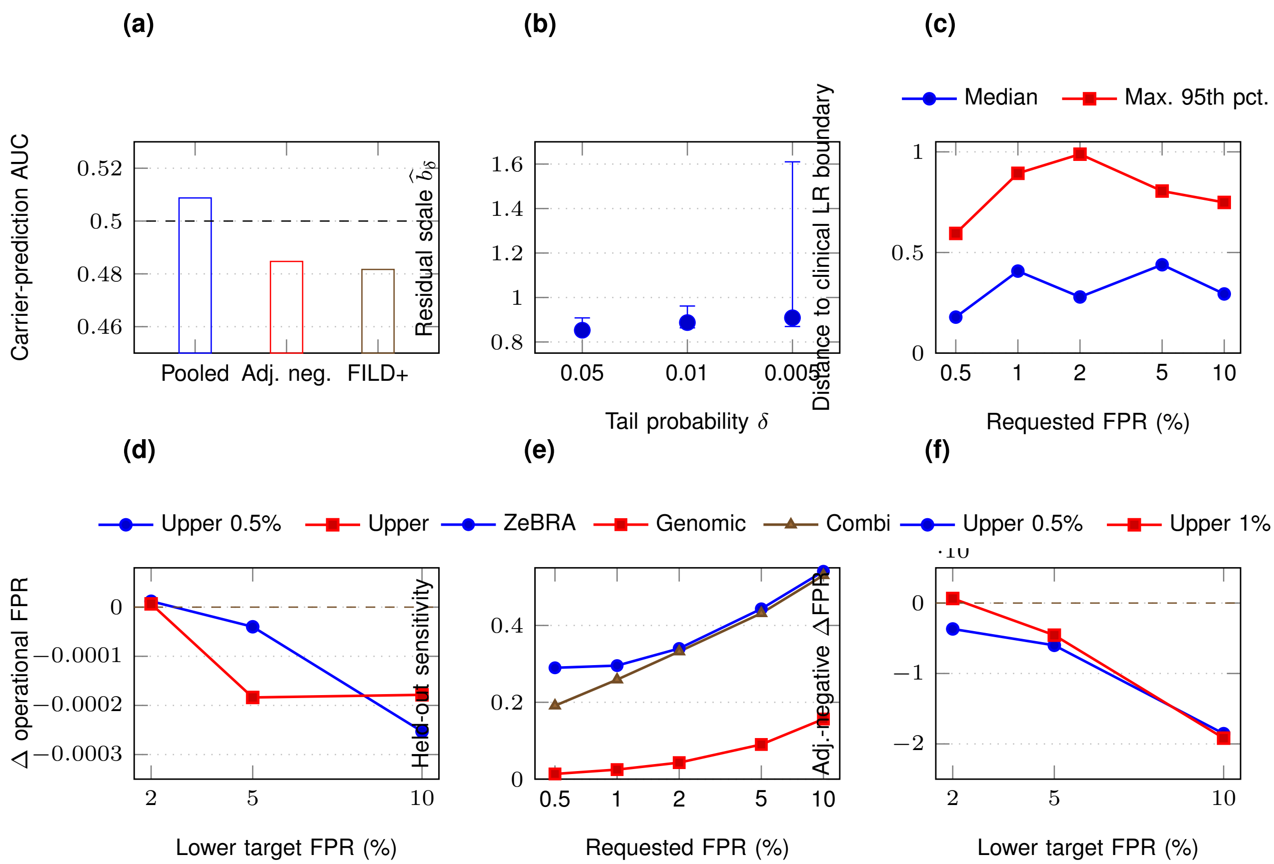


Fig. S1: **Supplementary FILD/FILA robustness and secondary diagnostics.** (a) ZeBRA predicts *MUC5B* carrier status at chance level in pooled and disease-stratified analyses, showing that the clinical score does not reconstruct the focal genotype. (b) Empirical high-probability residual-genomic scales across the nine prespecified direct-LR estimators; the 95% and 99% scales are stable while the 99.5% tail is more estimator sensitive. (c) Genomic-induced decision changes remain localized near the clinical LR boundary. (d) Operational FPR differences for the matched-FPR rescue rules are near zero. (e) Held-out sensitivity at stringent requested FPRs complements the LR+ comparison in main Fig. 1g. (f) The explicitly adjudicated-negative rescue diagnostic is directionally favorable for most wider rescue bands but is underpowered because the adjudicated-negative subset is small.